MILITARY SPECIFICATION

PUMPS, CENTRIFUGAL: PORTABLE, SELF-PRIMING, GASOLINE-ENGINE-DRIVEN (NONCLOGGING, TRASH-HANDLING, CONTRACTOR TYPE)

1. SCOPE

1.1 Scope. This specification covers portable, self-priming, gasoline-engine-driven, single stage, nonclogging, medium-head, centrifugal pumps designed to handle trash-laden water.

1.2 Classification. The pumps shall be of the following sizes and classes, as specified (see 6.2):

Size 19MT - 3-inch; 19,000 g.p.h.
Size 33MT - 4-inch; 33,000 g.p.h.

Class 1 - Base-mounted.
Class 2 - Wheel-mounted, two-wheel, manually-propelled, 4.00x8 pneumatic, wheelbarrow tires.
Class 2A - Wheel-mounted, two-wheel, vehicular-towed, 5.0-15 (minimum) pneumatic, automotive tires.
Class 3 - Skid-mounted.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-V-173 - Varnish, Moisture- and Fungus-Resistant, for the Treatment of Communications, Electronic and Associated Electrical Equipment.
MIL-P-514 - Plates, Identification, Instruction and Marking, Blank.
MIL-T-704 - Treatment and Painting of Materiel.

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MIL-F-3541 - Fittings, Lubrication.
MIL-P-10603 - Pumps and Pumping Units, Centrifugal, Power-Driven, for Water, Packaging of.
MIL-E-11275 - Engines, Gasoline, Industrial Type, General Specification for.

STANDARDS

Military

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-130 - Identification Marking of U.S. Military Property.
MIL-STD-461 - Electromagnetic Interference Characteristics Requirements for Equipment.
MIL-STD-462 - Electromagnetic Interference Characteristics Measurements of.
MIL-STD-1400 - Engines, Gasoline or Diesel, Methods of Test.

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM)

A48 - Gray Iron Castings.
A126 - Gray Iron Castings for Valves, Flanges, and Pipe Fittings.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race St., Philadelphia, Pa. 19103.)

Associated General Contractors of America (AGC)

Contractors Pump Standards.

(Application for copies should be addressed to the Contractors Pump Bureau, Associated General Contractors of America, Inc., Washington, D. C. 20006.)

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Hydraulic Institute (HI)

Hydraulic Institute Standards.

(Application for copies should be addressed to the Hydraulic Institute, 122 East 42nd St., New York, N.Y. 10017.)

American Trucking Associations, Inc. (ATA)

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., 1616 P Street, N.W., Washington, D.C. 20036.)

Uniform Classification Committee (UCC)

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, 202 Union Station, Chicago, Ill. 60606.)

Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.

3. REQUIREMENTS

3.1 Description. Each unit shall consist essentially of an integrally-self-priming centrifugal pump; a direct-connected gasoline engine; and a common base, skid, or wheel mounting. The pump capacity in gallons per hour (g.p.h.) shown under 1.2 to identify pump sizes shall be interpreted as the nominal capacity. The actual capacity shall be as further defined in the detailed requirements of this specification. The pumps shall be equipped with nonclogging, semi-open or open, two-vane, trash-type impellers and readily opened cleanout ports or covers.

3.2 First article. When specified (see 6.2), the contractor shall furnish a sample pump for first article inspection. This inspection shall be as specified in 4.2.1 and shall be conducted prior to the submission for acceptance of the remainder of the pumps to be furnished under the contract. The Government reserves the right to observe and, at the discretion of the procuring agency, to direct or supervise the first article inspection. The sample pump may be a standard production item from the contractor's current inventory provided the unit meets

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the requirements of this specification and is representative of the design, construction, and manufacturing techniques applicable to the remaining units to be furnished under the contract. Following acceptance by the Government, the sample may be delivered as the last unit of supply under the contract. Approval of the sample pump on the basis of the specified inspection shall not relieve the contractor of his obligation to meet the requirements of this specification for the remaining units to be furnished under the contract.

3.3 Standard product. The pump components of each unit shall be a standard commercial product of the pump manufacturer which meets or exceeds the requirements of this specification. Unless military engines are specified (see 3.9.5), the engine shall be a standard commercial model of the engine manufacturer except for any optional engine accessories specified herein, not offered as standard equipment by the engine supplier. The completely assembled pumping unit shall reflect overall the standard commercial practice of the contractor except for any modifications or deviations necessary to meet the requirements herein for the skid-, base-, or wheel-mountings.

3.4 Interchangeability. All pumping units of the same classification furnished under a specific contract shall consist of like models or sizes of pumps, engines, and other components to insure interchangeability of replacement parts.

3.5 Material. Material for pumps and other components shall be as specified herein. When material for a part is not definitely specified, the material shall be of the type normally used by the manufacturer. A class or grade of material having mechanical and physical properties equal to or better than the grade or class specified herein may be substituted, subject to approval of the procuring agency. Materials shall be free of defects or imperfections that would contribute to failure of the pumping unit or would materially reduce the usability and performance of the unit.

3.5.1 Iron castings. Gray cast iron used for pump casings, impellers, and wear plates shall meet the requirements of ASTM A48, class No. 25 or better; or of ASTM A126, class B (semisteel) or better. Ductile (nodular) iron or malleable iron castings may be used in lieu of gray iron castings provided the surface has a Brinell hardness number (BHN) of not less than 160.

3.5.2 Steel. Steel used for such components as pump shafts, impellers, wear plates, skids, bases, and diffusers shall be of a type and grade normally used by the supplier on his standard commercial units.

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3.6 Performance.

3.6.1 Capacity. The pumps shall meet the minimum capacity requirements specified in table I. The capacity requirements shall be attainable without exceeding the continuous horsepower rating of the engine as defined herein. Pump capacities at the specified heads and static suction lifts shall be based on pumping fresh, 68° F. water (62.3 pounds per cubic foot) at sea level (barometric pressure of 29.92 inches of mercury) using a suction hose or pipe equal in nominal size to the pump connections and five feet longer than the static suction lift.

TABLE I. Minimum pump capacities

Pump size	Total head including friction (feet)	Height of pump above water (feet) 10	15	20	25
19MT (3-inch)	25	310	265	205	115
	30	310	265	205	115
	40	290	262	205	113
	50	260	250	200	107
	60	215	215	190	100
	70	167	167	167	90
	80	117	117	117	78
	90	72	72	72	60
	100	22	22	22	22
33MT (4-inch)	30	550	460	360	240
	40	540	450	350	237
	50	515	437	337	230
	60	450	415	320	220
	70	360	360	300	209
	80	270	270	262	190
	90	180	180	180	160
	100	90	90	90	90

3.6.2 Priming. After initial filling with priming fluid, the pumps shall be capable, on succeeding startups, of automatically self-priming at static suction lifts up to 25 feet under the standard conditions of fresh water temperature and atmospheric pressure specified herein for capacity determinations.

3.6.3 Solids handling. The pumps shall be capable of passing spherical solids of the applicable sizes listed in table II without clogging impeller or pump passages. The pumps shall also be capable of passing solids equal to at least ten percent by volume of the water being pumped.

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TABLE II. Pump and engine requirements

Component	Requirement	Size 19MT (3-inch)	Size 33MT (4-inch)
Pump	Solids handling capability, sphere diameter, inches	1 1/2	2
	Hydrostatic pressure for pump casing, p.s.i.g.	90	90
Engine	Displacement, cubic inches, minimum	22	105
	Continuous horsepower, minimum	7 1/4	20
	Operating speed, r.p.m.'s, maximum	3600	2500
	Number of cylinders	1	4
	Starting system	Manual; rope or crank	Manual, crank; or, when specified (see 6.2), 12-volt electric
	Fuel tank capacity, gallons, minimum	1 1/2	8

Note 1/: A 12-volt, dry-charged battery shall be furnished with each electrically-started unit.

3.6.4 Environmental conditions. The pump units shall be capable of starting and operating, not necessarily at rated capacity, in ambient temperatures up to 120° F. at sea level and 110° F. at an elevation of 5000 feet (barometric pressure of 24.9 ± 0.2 inches mercury). The units shall also be able to withstand extended periods of storage at temperatures to minus 25° F. without adverse effect on the subsequent operability or efficiency of the units.

3.7 Pumps. The pump component shall be of the self-priming, radial flow, single stage, solids-handling centrifugal-type referred to commercially as a contractor trash pump. Each pump shall consist essentially of a casing, an impeller, a shaft seal, a wearplate(s), inlet and discharge connections, and air separation devices or chambers. The impeller shaft shall be an extension of the engine

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crankshaft except for the size 33MT, 4-inch pump which may be furnished with a separate, bearing-mounted, impeller shaft flexible-coupled to the engine crankshaft.

3.7.1 Casing. The casing shall be cast iron and shall be designed to withstand the hydrostatic test pressure specified in table II. The casing shall be equipped with a priming port and means for ready drainage of the casing. A standard steel pipe tee and plug installed on the discharge connection will be acceptable as a priming port. The casing shall be designed or integrally equipped to prevent siphonage of priming fluid on shutdown. The pumps shall be equipped with renewable wearplates or replaceable volute sections, or both, so that proper clearances can be restored without the need for replacing the pump casing.

3.7.1.1 Cleanout port. The pump casing shall be furnished with a cleanout port to permit removal of rags or other debris which may clog the impeller and with a suction end cover or hatch to provide accessibility for inspection and removal of the impeller, volute section, and wearplate. The inside diameter of the cleanout port shall be not less than the nominal pipe size of the pump suction connection. The cleanout port cover shall be secured in a manner to permit removal by hand without the use of tools. Acceptable fasteners shall include wing nuts, L- or T-handle screw-and-yoke devices, or a combination of both. The number of fasteners required for securing the cleanout port cover shall be four or less. The suction end cover may serve as both the cleanout port and impeller access opening provided the cover is secured by a method specified herein for the cleanout port.

3.7.2 Impeller. The impeller shall be cast iron, semisteel, ductile (nodular) iron, or malleable iron and shall be of the semi-open or open, two-vane, nonclogging type. The impeller shall be cast in one piece, balanced, and keyed or screwed to the shaft in a manner to provide positive locking against any rotative loosening effect.

3.7.3 Bearings. Pumps with separate shafts coupled to the engine crankshaft shall be equipped with at least two sets of ball or roller, antifriction bearings to support the shaft. The bearings shall be rated to withstand continuous radial or axial loading due to inherent rotational forces and hydraulic imbalance. The bearings shall have adequate provisions for lubrication and shall be sealed against the entrance of dirt and water.

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3.7.4 Shaft seal. The shaft seal shall be of the mechanical type and shall be self-lubricated, oil-lubricated, or grease-lubricated. The seal shall consist essentially of a rotating element, a stationary element, a spring, and any additional elements common to particular seal designs such as a diaphragm, bellows, spring cage, or grease cup. Stationary elements shall be fabricated of a hard, wear-resistant ceramic material or of nickel-bearing cast iron. On self-lubricated seals, the rotating wearing surface shall be carbon or a material having similar low-friction properties. Springs shall be stainless steel. Grease cups shall be of the spring-loaded type designed to automatically feed lubricant to the seal as required.

3.7.5 Connections. The suction opening shall be of the same nominal pipe size as the discharge opening, the sizes to be as specified in 1.2. The openings shall be tapped or fitted to provide female, American Standard taper pipe (NPT) connections. Both connections shall be so located or fitted that a direct-connected length of straight pipe will extend in a horizontal direction. Both connections shall be so designed or located that the cleanout port and suction end-plate can be removed without disturbing connected hoses or piping.

3.7.6 Suction strainer. A suction strainer (end-guard) shall be furnished with each pump. The strainer shall be furnished with a threaded connection of the same nominal pipe size as the connections on the pump for which it is supplied. The strainer shall be designed to permit passage only of solids having a size the pump can handle without clogging.

3.8 Mountings. Pumps shall be base-mounted, skid-mounted, or wheel-mounted in accordance with the specified class. Wheel-mounted, class 2 units shall be designed to be manually propelled; wheel-mounted, class 2A units shall be designed for vehicular towing.

3.8.1 Base mounting. Class 1 base mountings shall consist of a flat steel plate or a steel plate or sheet formed to provide a raised horizontal center section with supporting vertical sections along each longitudinal edge. Base mountings consisting of a combination of steel channels, flat steel runners, or other structural steel shapes will also be acceptable. The base mounting shall be of sufficient thickness and rigidity to maintain the pump and engine in proper alinement under all conditions of operation and transport.

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The base mounting shall be designed primarily for permanent or semi-permanent locations and shall be equipped with holes suitably sized and located for hold-down bolts. Base mountings shall otherwise conform to the manufacturer's standard, industrial-type mounting.

3.8.2 Wheel mounting.

3.8.2.1 Manually propelled (class 2). Manually propelled, class 2 mountings shall consist essentially of a two-wheel, rubber-tired cart with a push bar or handle, and a supporting leg. Tires shall be of the pneumatic, wheelbarrow-type. Tires and wheels shall be commercially rated to withstand speeds up to 10 miles per hour when supporting the gross weight of the pumping unit.

3.8.2.2 Vehicular-towed (class 2A). Wheel mountings for class 2A pumps shall be designed for vehicular towing over graded gravel roads at speeds up to 20 miles per hour. The wheel mounting shall include a drawbar and lunette-type towing ring. The towing ring shall be fabricated of one-inch round steel stock and shall form a ring with a 3-inch ± 1/16-inch outside diameter. The mounting shall include two safety chains with links formed of stock not less than 5/16 inch in diameter. The safety chains shall be anchored to the drawbar and shall extend 24 inches beyond the towing ring. The loose end of each chain shall be fitted with a safety-type hook or a pin-type clevis. Class 2A mountings shall be equipped with one or more retractable landing legs located, as required, to prevent the unit from tipping when it is disconnected from the towing vehicle. The leg or legs shall also be capable of maintaining the unit in a stable, level attitude during operation.

3.8.3 Skid mounting. Skid mountings shall be furnished when class 3 pumps are specified. The skid mounting shall consist of a base mounting having longitudinal steel skids along each of the two longest edges of the base mounting. The skids shall be structural steel shapes, such as channels, I-beams, or box columns, and shall be of sufficient depth to provide at least a three-inch clearance between the ground and the underside of the base plate or cross members. The bearing surface of each skid shall be not less than two inches wide. Both ends of each skid shall be beveled or rounded to facilitate skidding. Both ends of each skid shall also be equipped with pulling attachments. A hole not less than two inches in diameter near the ends of each skid will be acceptable in lieu of pulling attachments. The pump and engine shall be mounted to the skid base in a manner to permit ready removal and replacement of either or both components. Skid mountings shall be designed

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and constructed to permit pulling (skidding) the units over dry, unpaved surfaces at speeds up to five miles per hour without damage to the skid or damage to or misalinement of any components mounted on the skid.

3.9 Engine. The engine shall be an industrial, air-cooled, four-cycle gasoline engine. The engine shall meet the applicable requirements of table II.

3.9.1 Horsepower. The engine, fully equipped with all accessories, shall be capable of driving the pump to deliver fresh water under standard conditions at rated capacity for all applicable conditions of head and lift specified herein without exceeding the continuous horsepower rating of the engine. The continuous horsepower ratings shall be established in accordance with recognized engine testing standards at not more than 80 percent of full throttle. The minimum continuous horsepower rating and the maximum operating speed at which the rating is to be established shall be as specified in table II.

3.9.2 Engine accessories. Each engine shall be equipped with a governor, a manual speed control, an exhaust muffler, an air cleaner, and fuel tank. Air-cooled engines shall also be equipped with a fan-flywheel having a protective screen or shroud. All gasoline engines shall be equipped with positive exhaust valve rotators and with stellite-faced exhaust valves and solid stellite valve-seat inserts. All other engine accessories supplied in commercial practice as standard equipment shall be furnished.

3.9.3 Starting systems. The starting system shall be a manual (rope or crank) or electric system in accordance with table II. Electric starting systems shall be 12-volt and shall include an electric cranking motor, charging generator or alternator, regulator, battery with cables, starter switch, and an ammeter. Gasoline engines with electric starting systems shall include a high-tension, distributor-type ignition system. A battery shall be included with each electrically-started unit and shall be shipped charged and dry.

3.9.4 Hour meter. When specified (see 6.2), four-cylinder engines shall be equipped with an hour meter. The meters shall either be electrical or mechanical and shall be capable of registering at least 9999 hours of operating time.

3.9.5 Military engine. When specified (see 6.2), gasoline engines shall meet the requirements of MIL-E-11275 and shall be listed or approved for listing on the associated Qualified Products List or

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any amendments thereto. The engines shall be goup I or group II of MIL-E-11275, as applicable. Military engines shall meet the other requirements of table II, including displacement, horsepower, and operating speeds. The continuous horsepower rating shall be established from the qualification tests.

3.9.5.1 Group I. Group I engines shall be industrial, air-cooled, 4-cycle type gasoline engines having a net continuous horsepower rating of less than 10 horsepower. In accordance with MIL-E-11275, the engine shall be furnished either with Military standard parts or commercial parts at the option of the engine manufacturer. However, the engine shall be furnished with parts identical to those used during the qualification tests. Group I engines shall be equipped with all applicable commercial accessories specified in 3.9.2 in addition to complying with MIL-E-11275 for integrally furnished engine components such as the carburetor, ignition system, bearings, and valves.

3.9.5.2 Group II. Group II engines shall be 4-cylinder, 4-cycle, with a net continuous horsepower rating over 10 horsepower. The engines shall be equipped with the high-mortality parts required under MIL-E-11275. The engines shall also be equipped with the applicable accessories specified in 3.9.2 except that the accessories shall conform to MIL-E-11275 if covered by the provisions therein. Additional engine-system components and accessories required under the class A identification of MIL-E-11275 shall be furnished. Engine-system components and accessories listed under class B will not be required unless otherwise specified in the contract (see 6.2).

3.10 Lubrication. Except as otherwise required for level A preservation (see 5.1.1), the pumping units shall be serviced prior to shipment with grease and oils of the types and grades recommended by each component manufacturer. Lubrication fittings shall conform to types I, II, III, or IV of MIL-F-3541 and shall be located in accessible protected positions. When use of high pressure lubricating equipment (1000 p.s.i. and higher) will damage grease seals or other parts, a suitable warning plate shall be affixed to the unit in a conspicuous location. Although initially serviced with standard commercial lubricants, the units shall be capable of operating efficiently and satisfactorily without deleterious effects when using standard military lubricants (see 6.3).

3.11 Treatment and painting. The pump mounting plus the surfaces of the pump and engine normally painted shall be cleaned, treated, and painted in accordance with MIL-T-704, type A. The color shall be as specified (see 6.2).

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3.12 Lifting attachments. All pumping units covered by this specification shall be equipped with lifting eyes or lifting bails. The lifting devices shall be designed and located to permit the units to be lifted in the normal travel position. Each lifting device shall be designed to withstand a loading equal to 2 1/2 times the normal static load imposed when the pump is suspended in a stationary position.

3.13 Tiedown devices. When specified (see 6.2), the units shall be equipped with permanently affixed devices to permit tiedown of the fully assembled pumping units to the floor or deck of the transporting medium. The tiedown devices shall be of such design, location, and number as to enable the unit to be anchored without shifting in any direction.

3.14 Identification marking. The pumps shall be marked for identification in accordance with MIL-STD-130. The marking shall be applied to the pumps on a plate conforming to MIL-P-514. Plates shall be furnished by the contractor unless procurement documents indicate the plates will be Government-furnished. Contractor-furnished plates shall be an aluminum or copper-base alloy plate conforming to type I, style 1, composition A or C of MIL-P-514. The contractor shall apply the information required under MIL-STD-130 to the plates, whether contractor- or Government-furnished, using a method of application specified in MIL-P-514. Plates shall be attached to the equipment by bolts, screws, or rivets in a conspicuous location affording protection against damage or disfigurement during normal use of the pumps.

3.15 Electromagnetic interference control. When specified (see 6.2), the units shall meet the electromagnetic interference control requirements and test limits for class IIIC equipment as specified in MIL-STD-461.

3.16 Fungus resistance. When specified (see 6.2), engine electrical connections, including terminal and circuit connections, components, and circuit elements shall be coated with varnish conforming to MIL-V-173 except that:

a. Components and elements inherently inert to fungi or hermetically sealed need not be treated.

b. Components and elements whose operation will be adversely affected by the application of varnish shall not be treated.

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c. One percent of copper 8-quinolinolate (by weight based on the nonvolatile content of the varnish) may, at the supplier's option, be substituted for the salicylanilide specified in MIL-V-173, provided that the varnish conforms to MIL-V-173 in all other respects.

3.17 Technical publications. When specified (see 6.2), such publications as operating sheets or manuals, maintenance and service instructions, and replacements parts lists shall be furnished. The specific publications to be furnished, the contents thereof, the number of copies required, and the distribution of copies shall be as specified in the contract (see 6.2), or, when applicable, shall be determined in accordance with provisioning procedures of the contract. When specific requirements for technical publications are not included in the contract, the manufacturer shall furnish with each unit a copy of the publication(s) normally supplied with commercial consignments.

3.18 Spare parts. When specified (see 6.2), spare parts shall be furnished and shipped with each unit. The spare parts required and the quantity thereof shall be as specified in the contract, or, when applicable, shall be determined in accordance with the provisioning procedures of the contract.

3.19 Workmanship. The unit shall be free of defects such as incomplete welds, cracks, or other defects that could impair the operation or serviceability of the unit.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection.
b. Quality conformance inspection.
   1. In-process examination.
   2. End-item inspection.

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4.2.1 First article inspection. First article inspection shall be performed on one unit when a first article sample is required (see 3.2). This inspection shall include the examinations of 4.4.1 and 4.4.2 and the tests of 4.5.1 and, when applicable, 4.5.1.6.

4.2.2 Quality conformance inspection. Quality conformance inspection shall be as follows:

a. In-process inspection shall be performed on each unit to be offered for acceptance by the Government and shall include the examination specified in 4.4.1.

b. End-item inspection shall be performed on samples selected in accordance with 4.3. This inspection shall include the examination of 4.4.2 and the tests of 4.5.2.

4.3 Sampling.

4.3.1 Sampling for examination. A random sample of pumps shall be selected from each lot offered to the Government in accordance with inspection level II of MIL-STD-105. The acceptable quality levels (AQL's) shall be 2.5 percent defective for major defects and 4.0 percent defective for minor defects.

4.3.2 Sampling for tests. A random sample of pumps shall be selected from each lot offered the Government in accordance with inspection level S-4 of MIL-STD-105. The AQL shall be 2.5 percent defective.

4.3.3 Lot. A lot shall consist of all units of the same size and class offered for delivery to the Government at one time under a specific contract. A sample unit shall consist of one completely assembled pumping unit.

4.4 Examination.

4.4.1 In-process examination. Pumping units furnished under this specification shall be examined during production in accordance with the manufacturer's established quality control program. The quality control methods and procedures shall assure compliance with all requirements of this specification which can be checked visually and with those elements of the manufacturer's drawings or specifications normally verified during the inspection process. The requirement for in-process examination shall apply to both the pump component

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and the engine component. The pump manufacturer shall be responsible for verifying, when requested, that suitable in-process examination has been performed during production of the engine component. The pump manufacturer's quality control program shall be subject to surveillance by the procuring agency or designated representative thereof.

4.4.2 End item inspection. Sample units selected in accordance with 4.3.1 shall be examined to verify compliance with the nonoperational requirements of this specification. This examination shall verify that the units are complete and that all components, accessories, and controls are furnished as specified for the applicable size and class and are in conformance with applicable requirements. This inspection shall also confirm that such requirements as interchangeability, material, painting, lubrication, lifting devices, marking, and similar provisions are complied with. Defects shall be classified in accordance with MIL-STD-105 and if the number of defective units is equal to or greater than the rejection number applicable to the specified AQL, the lot represented by the sample units shall be rejected.

4.5 Tests. Prior to the tests specified herein which require the units be operated, the units shall be serviced with lubricants recommended by the manufacturer for use in the ambient temperature at the test location. Batteries used for test purposes on units equipped with electric starting systems shall be supplied by the manufacturer for use in lieu of the dry-charged battery that is to be shipped with the unit.

4.5.1 First article tests.

4.5.1.1 Capacity test. The unit shall be tested to verify compliance with the head-capacity requirements of 3.6.1. The tests shall be performed in accordance with the provisions of the Hydraulic Institute standards cited in the AGC Contractors Pump Standards. Failure of the pump to meet any head-capacity requirement shall constitute failure of the test.

4.5.1.2 Priming test. The unit shall be tested to verify compliance with the priming requirement of 3.6.2. The test conditions shall be as specified in 3.6.2. The pump shall be operated at a speed not exceeding the engine manufacturer's recommended maximum operating speed or at the maximum operating speed specified in table II, whichever is less.

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4.5.1.3 Environmental tests. When specified (see 6.2), the first article shall be tested to verify compliance with the environmental requirements of 3.6.4. The tests shall be conducted in accordance with the applicable test methods of MIL-STD-1400.

4.5.1.4 Mobility tests. Skid-mounted units shall be towed at five miles per hour in a series of S-curves for a minimum distance of 300 feet. Class 2A units shall be towed at speed up to 20 miles per hour over graded gravel roads for a distance of at least 15 miles. Evidence of structural deformation or failure of the mounting or damage or misalinement affecting serviceability either of the engine or pump shall constitute failure of the mobility test.

4.5.1.5 Engine horsepower. For commercial engines, the pump manufacturer shall make available, when requested, performance curves showing horsepower ratings and operating speeds for the engine being furnished. The curves shall be based on dynamometer tests certified by the engine manufacturer.

4.5.1.6 Electromagnetic interference control tests. When electromagnetic interference control is specified (see 3.14), the first article shall be tested by the supplier in accordance with the applicable test method of MIL-STD-462 to verify compliance with the requirements of MIL-STD-461 for class IIIc equipment. The supplier shall furnish the contracting officer, for approval, the interference control plan, the EMI/EMC test plan and the test report required in MIL-STD-461. Upon approval of the test report by the contracting officer and provided all other requirements of the specification are met, the first article shall be used as a model for all production units.

4.5.2 Quality conformance tests.

4.5.2.1 Operational tests. Sample units shall be run-in for a period of at least one hour to verify satisfactory operability of the units. The following items shall be checked before or during the tests:

a. Alinement of pump and engine.
b. Operation of mechanical seals and lubrication systems.
c. Freedom of liquid passages.
d. Operability of engine governor and manual speed control.
e. Proper functioning and calibration of engine meters and gages.
f. Any condition evidenced by excessive temperature, vibration, or noise.

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4.5.2.2 Lifting devices. The static load as specified in 3.12 for lifting attachments shall be applied to the attachment when mounted on its structural member. Weld failure or permanent deformation of the attachment or the structural member shall constitute failure of the test.

4.5.3 Material tests. Material tests shall be performed by the material supplier in accordance with the sampling procedures and test methods specified in the referenced specifications and standards. Pump manufacturers will not be required to perform material tests but shall be responsible for the receipt, identification, and proper application of material. The pump manufacturer shall maintain a general quality certification of the materials used and, when requested, shall make available to the contracting officer or his authorized representative the quality certification applicable to these materials.

4.6 Inspection of preparation for delivery. The preservation, packaging, packing and marking of the pumping units shall be examined to determine compliance with the requirements of section 5 of this specification.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C as required (see 6.2).

5.1.1 Level A. The pumps shall be preserved and packaged in accordance with the applicable level A requirements of MIL-P-10603.

5.1.2 Level C. The pumps shall be preserved and packaged in accordance with the supplier's standard practice.

5.2 Packing. Packing shall be level A, B, or C as required (see 6.2).

5.2.1 Level A. The pumps shall be packed in accordance with the applicable level A requirements of MIL-P-10603.

5.2.2 Level B. The pumps shall be packed in accordance with the applicable level B requirements of MIL-P-10603.

5.2.3 Level C. The pumps shall be packed in a manner which will insure arrival at destination in satisfactory condition and be acceptable to the carrier at lowest rates. Containers and packing shall comply with Uniform Freight Classification rules or National Motor Freight Classification rules.

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5.3 Marking. In addition to any special marking required by the contract or order, interior packages, and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. Pumps covered by this specification are intended primarily for pumping trash-laden water and slurries. The nomenclature and designations employed herein for pump sizes and classes have been developed to effect compatability with corresponding classifications used for similar contractor pumps covered by MIL-P-17861.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Size and class required (see 1.2).

c. When a sample unit for first article inspection is required (see 3.2).

d. When electric starting systems in lieu of hand cranks shall be furnished on size 33MT (see table II and 3.9.3).

e. When an hour-meter is required on four-cylinder engines (see 3.9.4).

f. When Military engines conforming to MIL-E-11275 are required, and class B accessories, if required for group II engines (see 3.9.5, 3.9.5.2, and MIL-E-11275).

g. Color required (see 3.11).

h. When tiedown devices are required on skid-mounted and automotive-wheel-mounted units (see 3.13).

i. When electromagnetic interference control is required (see 3.15).

j. When treatment for fungus resistance is required (see 3.16).

k. When technical publications are required and the detailed requirements thereof (see 3.17).

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l. When spare parts are required and the parts to be furnished or the specific details of the provisioning procedures applicable to furnishing of spare parts (see 3.18).

m. If first article inspection is required, whether first article shall be subjected to environmental tests (see 4.5.1.3).

n. Level of preservation and packaging and level of packing required (see 5.1 and 5.2).

6.3 Lubricants. If deemed necessary, procurement documents should specify that the contractor furnish a list of lubricants recommended for the engine and for grease-lubricated pumps. This list should include the type, grade, manufacturer's name and, when applicable and available, the appropriate Military specification number. To assist manufacturers in identifying suitable Military lubricants, the contracting officer should be prepared to supply to the contractor, when requested, a list of applicable lubricants as contained in the Federal Supply Catalog, Department of Defense Section, Identification List C9100-IL for Federal Supply Group 91 and pertinent specifications. The finalized and approved list of lubricants should then be included in the technical publications furnished with the units.

6.4 Military engines. When available in the horsepower ratings and displacement required, Military standard engines conforming to MIL-E-11275 should be specified if the equipment is to be procured in quantity for stock and issue and spare parts support will be provided by the Military supply system. If the pump units are to be procured locally as a single item or in small quantities for immediate use in base maintenance or construction work, commercial engines can be specified provided parts support will be provided by local commercial sources.

6.5 First article inspection. If first article inspection is desired or required, appropriate provisions should be included in the contract to allow the contracting officer to waive completely or in part the first article inspection and reduce the bid price accordingly if the successful bidder has previously performed any or all of the first article tests in recent procurements. A partial waiver may be especially applicable to specific models of gasoline engines which have previously passed electromagnetic interference control tests and the required EMI/EMC plan and report has been prepared and submitted. When the contract allows for partial or total waiver of first article

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inspection, the contract should require that the successful bidder seeking such a waiver submits satisfactory evidence, including test and inspection reports that the inspection has been performed, and such additional information as the contract identification, the procuring activity, the date, and other pertinent data deemed necessary by the contracting officer.

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